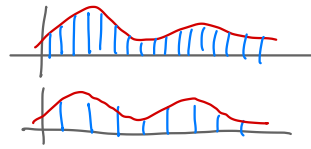
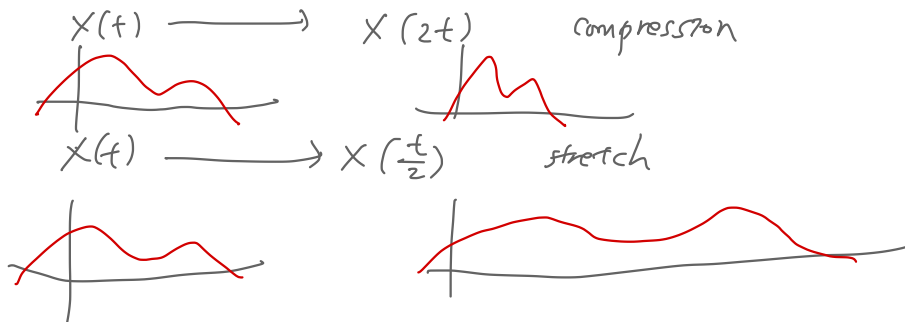


Rate Conversion:

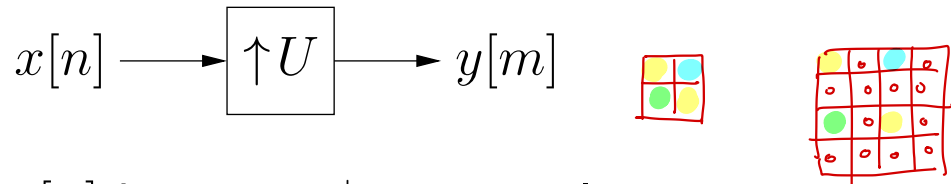


- Sometimes we would like to change the effective sampling rate of a signal *after* it has been sampled.
- To do this, we combine one of the elementary operations
 - *downsampling* (throwing away samples)
 - *upsampling* (inserting zero-valued samples)with discrete-time lowpass filtering in order to perform
 - *decimation* (decreasing effective sampling rate) and
 - *interpolation* (increasing effective sampling rate),respectively. These last two procedures change the effective sampling rate by an integer factor.
- When we want to change the effective sampling rate by a rational factor $\frac{U}{D}$, we combine interpolation and decimation into what is known as *resampling*.

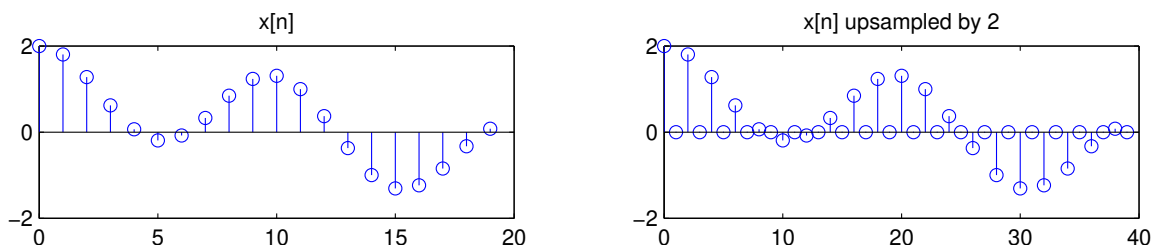


Upsampling:

- We represent upsampling by the block diagram



- To upsample $x[n]$ by $U \in \mathbb{Z}^+$, we simply insert $U - 1$ zeros between each sample in $x[n]$.

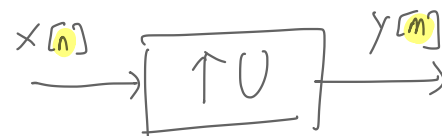


- In the time domain, upsampling can be written as

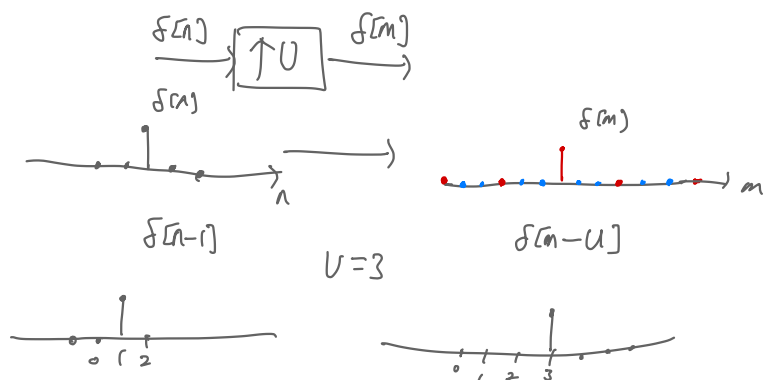
$$y[m] = \begin{cases} x[\frac{m}{U}] & \text{if } \frac{m}{U} \in \mathbb{Z} \text{ OR } m \text{ is a multiple of } U \\ 0 & \text{else} \end{cases}$$

- Is upsampling

- linear? **yes**
- time-invariant? **no**
- causal? ~~yes~~
- stable? **yes**



$$\max_m |y[m]| = \max_n |x[n]|$$



$$\{f[n-1]\} \neq \{f[n-U]\} \neq \{f[n-1]\} * \{f[n]\}$$

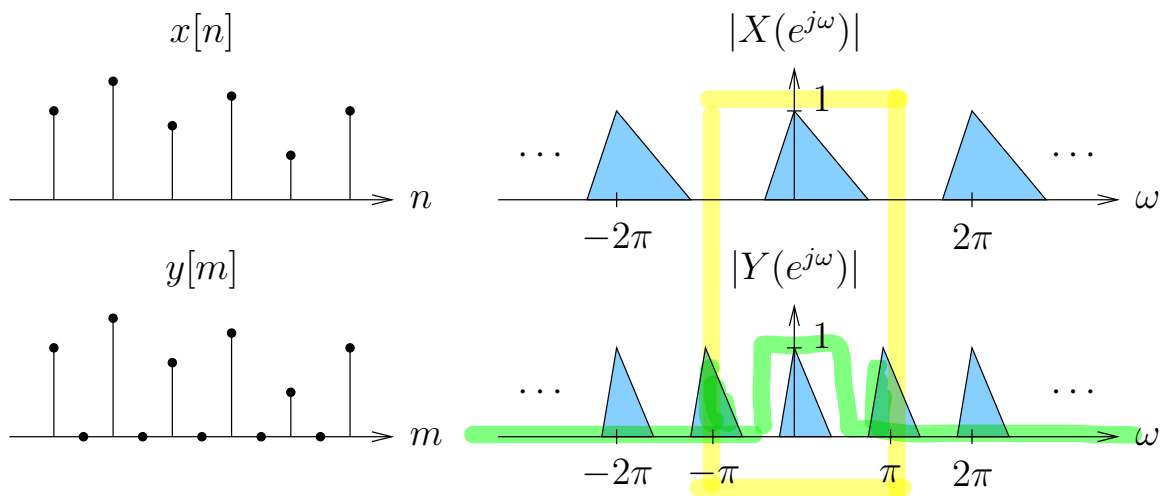
Upsampling (cont.):

In the frequency domain, upsampling can be written as

$$\begin{aligned}
 Y(e^{j\omega}) &= \sum_{m=-\infty}^{\infty} y[m] e^{-j\omega m} \\
 &= \sum_{m: \frac{m}{U} \in \mathbb{Z}} x\left[\frac{m}{U}\right] e^{-j\omega m} \quad (\text{e.g., } m = \dots, 0, U, 2U, \dots) \\
 &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega U n} \\
 &= X(e^{j\omega U})
 \end{aligned}$$

$X(e^{j\omega}) \rightarrow X(e^{j\omega U})$
 $U > 1$
 compression

For example, when $U = 2$,

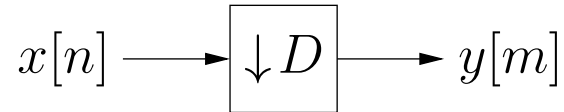


The spectrum is simply compressed by the integer factor U .

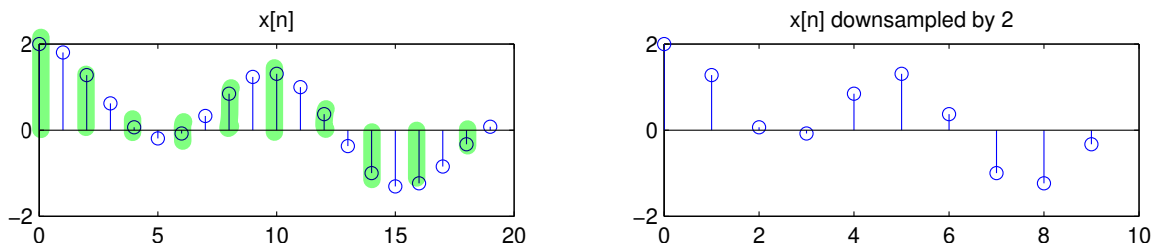
$$\frac{2\pi}{U} - \text{periodic}$$

Downsampling:

- We represent downsampling by the block diagram



- To downsample $x[n]$ by $D \in \mathbb{Z}^+$, we keep every D^{th} sample of $x[n]$ and discard the others.



- In the time domain, downsampling can be written as

$$y[m] = x[Dm]$$

- Is downsampling
 - linear? **yes**
 - time-invariant? **no**
 - causal? ~~yes~~ **no**
 - stable? **yes**

$$\max_m |y[m]| \leq \max_n |x[n]|$$

Downsampling (cont.):

discrete-time
impulse train

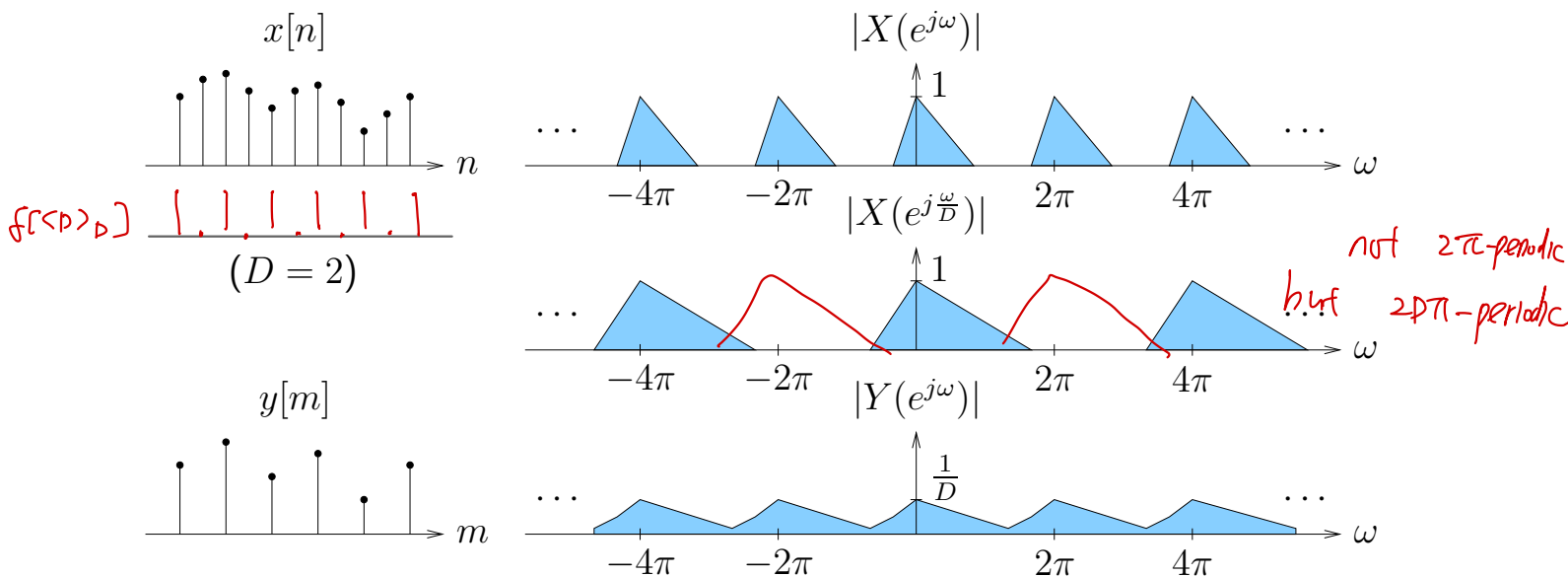
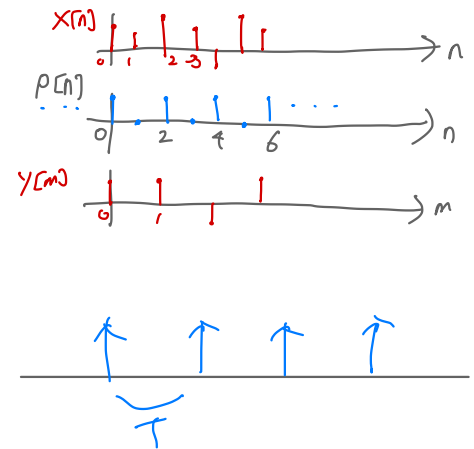
In the frequency domain, downsampling can be written as

$$Y(e^{j\omega}) = \sum_{m=-\infty}^{\infty} \underbrace{x[mD]}_{y[m]} e^{-j\omega m} = \sum_{n=-\infty}^{\infty} x[n] \underbrace{\delta[\langle n \rangle_D]}_{=1 \text{ iff } n \text{ is a mult of } D} e^{-j\omega \frac{n}{D}}$$

$$= \sum_{n=-\infty}^{\infty} x[n] \left(\frac{1}{D} \sum_{k=0}^{D-1} e^{j\frac{2\pi}{D}kn} \right) e^{-j\omega \frac{n}{D}} \quad (\text{next page!})$$

$$= \frac{1}{D} \sum_{k=0}^{D-1} \sum_{n=-\infty}^{\infty} x[n] e^{-j(\frac{\omega - 2\pi k}{D})n}$$

$$= \frac{1}{D} \sum_{k=0}^{D-1} X(e^{j\frac{\omega - 2\pi k}{D}})$$



Spectrum is expanded by D , scaled by $\frac{1}{D}$, and made periodic.

Useful Facts:

$$\langle n \rangle_D$$

modulo D

$$3 = 3 \cdot 1 + 0$$

$$4 = 3 \cdot 1 + 1$$

Why is $\delta[\langle n \rangle_D] = 1$ iff n is an integer mult of D ?

If $D = 3$,

n	-3	-2	-1	0	1	2	3	4	5
$\langle n \rangle_D$	0	1	2	0	1	2	0	1	2
$\delta[\langle n \rangle_D]$	1	0	0	1	0	0	1	0	0

Why is $\frac{1}{D} \sum_{k=0}^{D-1} e^{j \frac{2\pi}{D} kn} = \delta[\langle n \rangle_D]$? (Discrete Poisson sum.)

Draw pictures for $D = 3$ and the following four cases:

$$n = 0, 1, 2, 3.$$

case 1) $\langle n \rangle_D = 0 \Leftrightarrow n = Dm \quad m \in \mathbb{Z}$

$$e^{j \frac{2\pi}{D} kn} = e^{j \frac{2\pi}{D} k D m} = e^{j 2\pi km} = 1$$

$$\text{LHS} = 1$$

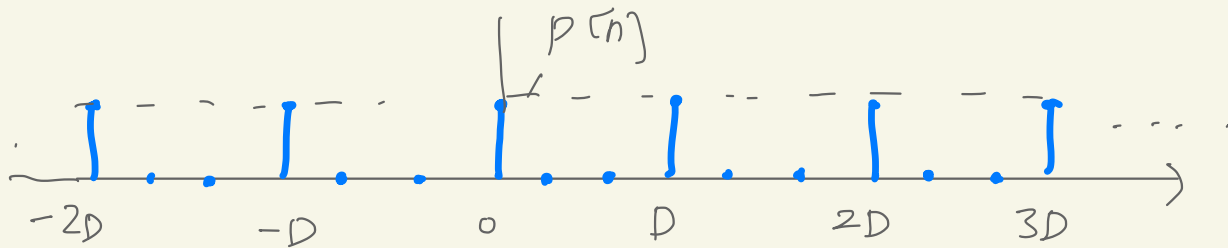
case 2) $\langle n \rangle_D \neq 0 \Leftrightarrow e^{j \frac{2\pi}{D} n} \neq 1$

$$\sum_{k=0}^{D-1} \underbrace{\left(e^{j \frac{2\pi}{D} n} \right)}_{\alpha}^k = \frac{1 - \alpha^D}{1 - \alpha} = \frac{1 - \cancel{e^{j \frac{2\pi}{D} n \cdot D}}}{1 - e^{j \frac{2\pi}{D} n}}$$

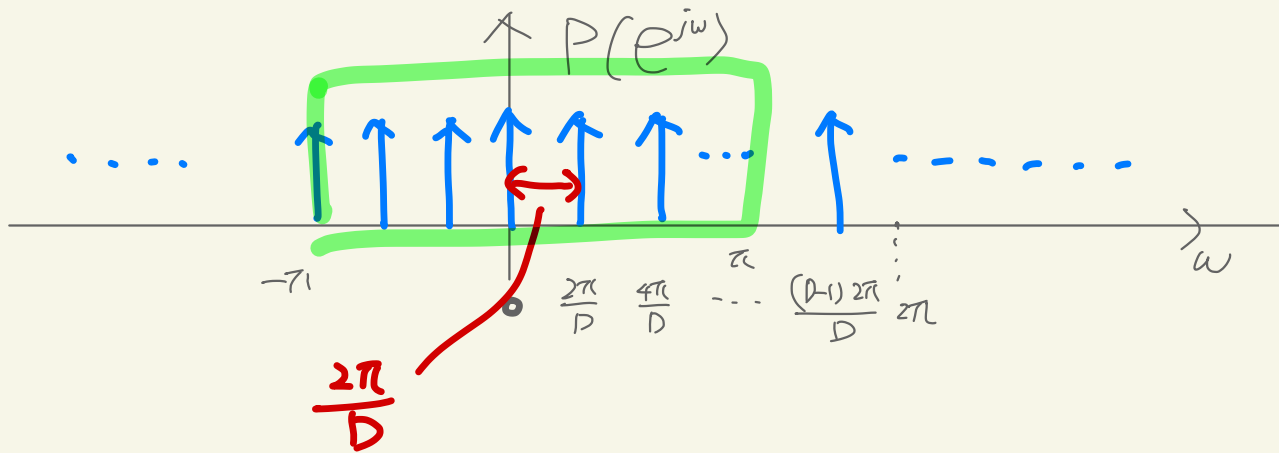
$$p[n] = \sum_{m=-\infty}^{\infty} \delta[n + Dm]$$

HW3 P4

$$P(e^{j\omega}) = \frac{2\pi}{D} \sum_{l=-\infty}^{\infty} \delta(\omega - \frac{2\pi}{D}l)$$

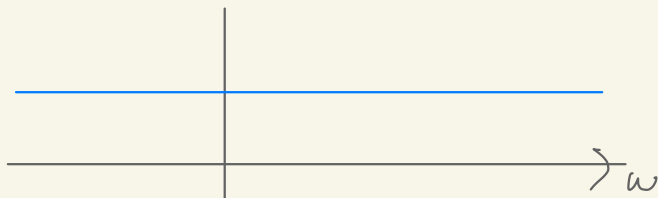
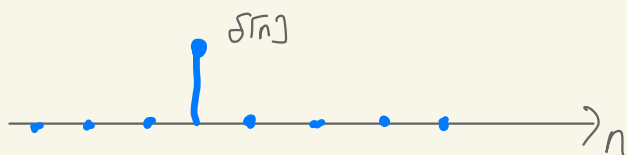


$p[n] = \delta[\langle n \rangle_D]$ discrete-time impulse train.



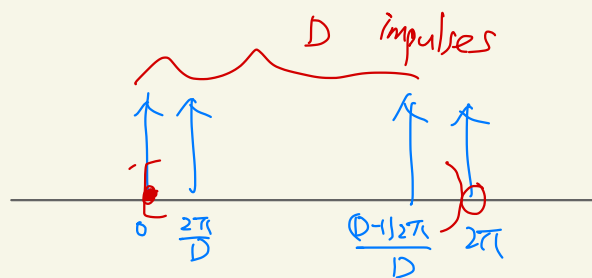
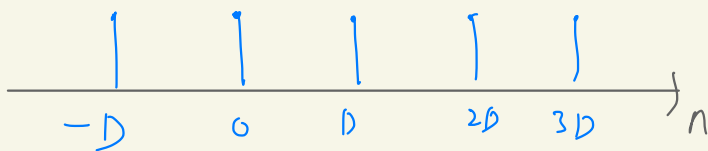
$$\delta[n] \xleftrightarrow{\text{DTFT}} 1$$

$$p[n] \longleftrightarrow \frac{2\pi}{D} \sum_{l=-\infty}^{\infty} \delta(\omega - \frac{2\pi}{D}l)$$



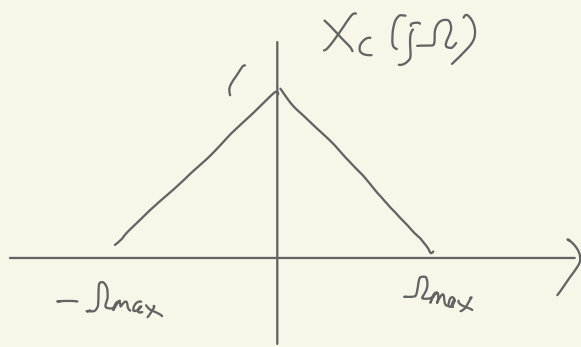
$$\delta[n] * p[n] = p[n] \xleftrightarrow{\text{DTFT}}$$

$$1 \cdot \frac{2\pi}{D} \sum_{l=-\infty}^{\infty} \delta(\omega - \frac{2\pi}{D}l)$$

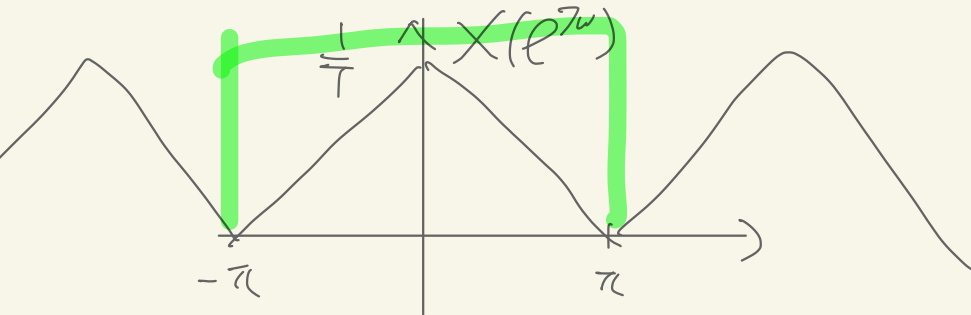


$$p[n] = \frac{1}{2\pi} \int_{[0, 2\pi)} \frac{2\pi}{D} \sum_{l=0}^{D-1} \delta(\omega - \frac{2\pi}{D}l) e^{j\omega n} d\omega$$

$$= \frac{1}{D} \sum_{l=0}^{D-1} e^{j\frac{2\pi}{D}ln}$$

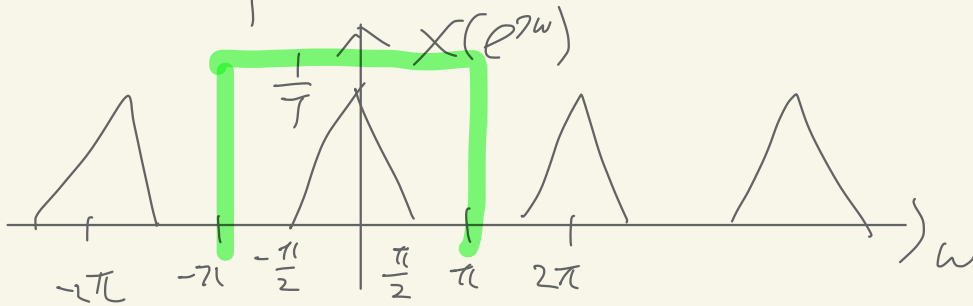


$$\frac{\pi}{T} = \Omega_{max}$$



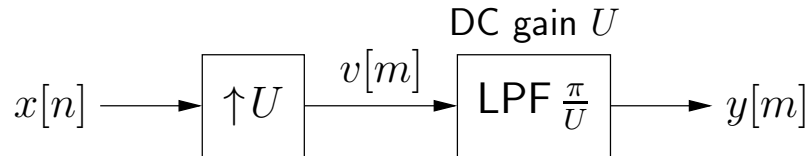
critically sampled.

$$\frac{\pi}{T} = 2\Omega_{max}$$



Interpolation:

- Interpolation goal: increase effective sampling rate from $\frac{1}{T}$ Hz to $\frac{U}{T}$ Hz. We can accomplish this in two steps:



- To understand the choice of LPF, recall that

$$X[n] = \mathcal{X}_c(nT) \Leftrightarrow X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega - 2\pi k}{T}\right)\right)$$

$$Y[n] = \mathcal{X}_c\left(\frac{nT}{U}\right) \Leftrightarrow Y(e^{j\omega}) = \frac{U}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega - 2\pi k}{T/U}\right)\right),$$

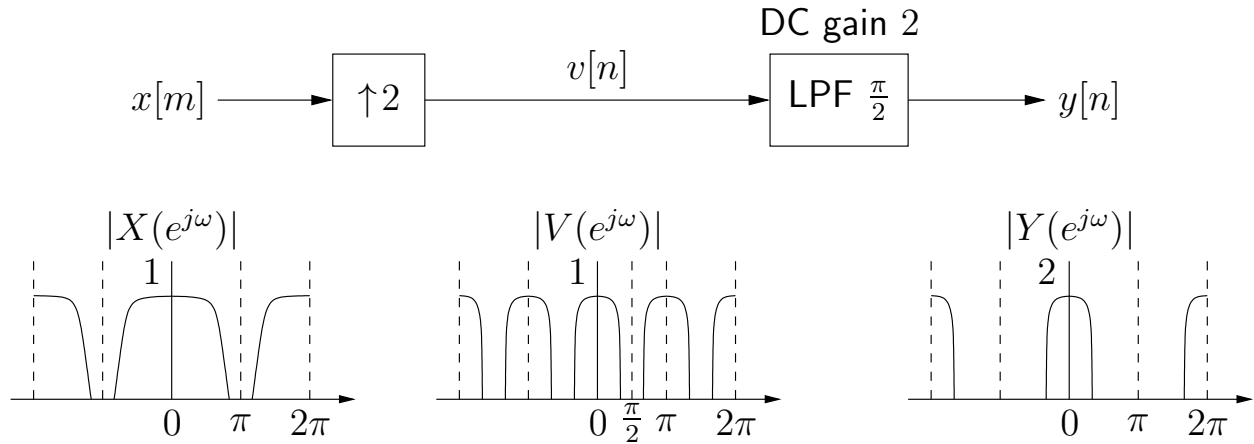
and notice that

$$\begin{aligned} V(e^{j\omega}) &= X(e^{j\omega U}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega U - 2\pi k}{T}\right)\right) \\ &= \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c\left(j\left(\frac{\omega - \frac{2\pi k}{U}}{T/U}\right)\right). \end{aligned}$$

We see that $V(e^{j\omega})$ has spectral copies at $\omega \in \{\frac{2\pi}{U}k\}_{\forall k}$ while $Y(e^{j\omega})$ has them at $\omega \in \{2\pi k\}_{\forall k}$. Thus, digital LPFing is needed to suppress the unwanted spectral copies (and scale by U).

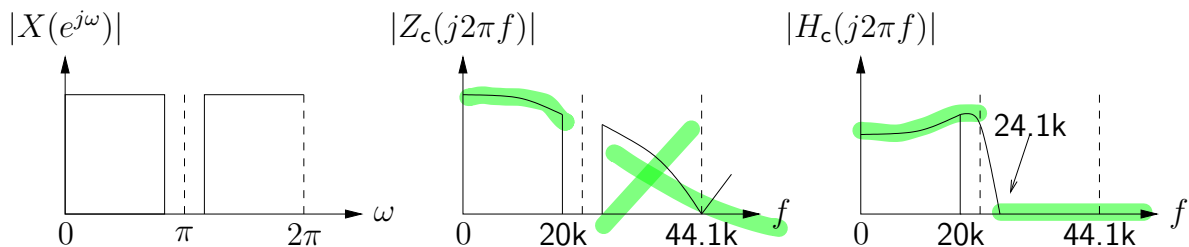
Interpolation (cont.):

- Interpolation example for $U = 2$:

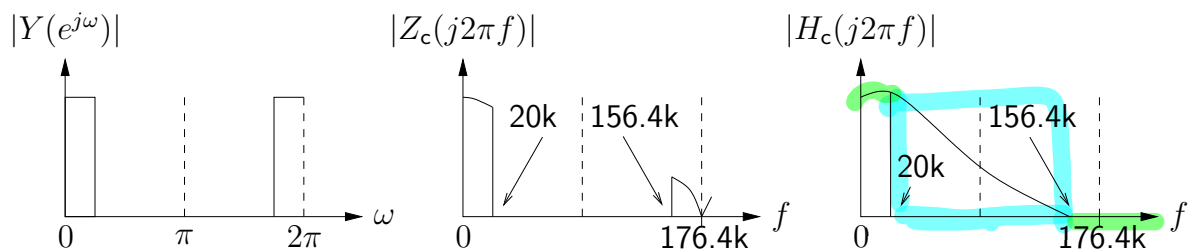


LPFing removes the copies centered at $\omega \in \{\pi k\}_{\forall \text{ odd } k}$.

- Application — CD-Audio D/A conversion:
 - ZOH reconstruction without oversampling:



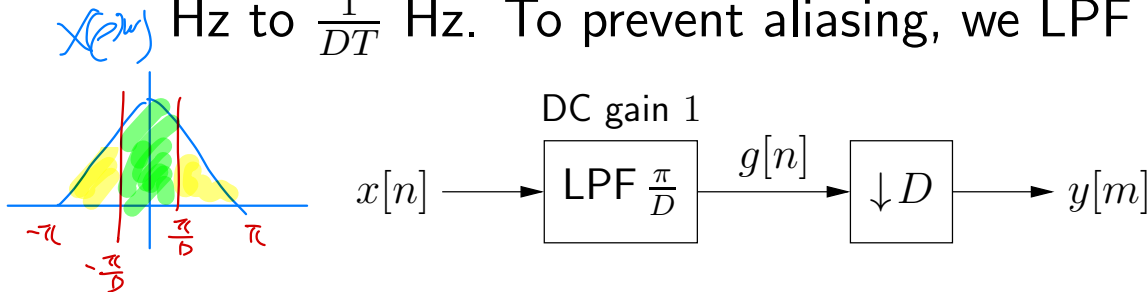
- ZOH reconstruction after $4\times$ oversampling:



Analog filter $H_c(j\Omega)$ becomes much cheaper!

Decimation:

- Decimation goal: decrease effective sampling rate from $\frac{1}{T}$ Hz to $\frac{1}{DT}$ Hz. To prevent aliasing, we LPF first:



- To understand the LPFing, say that $x[n] = x(nT)$ with

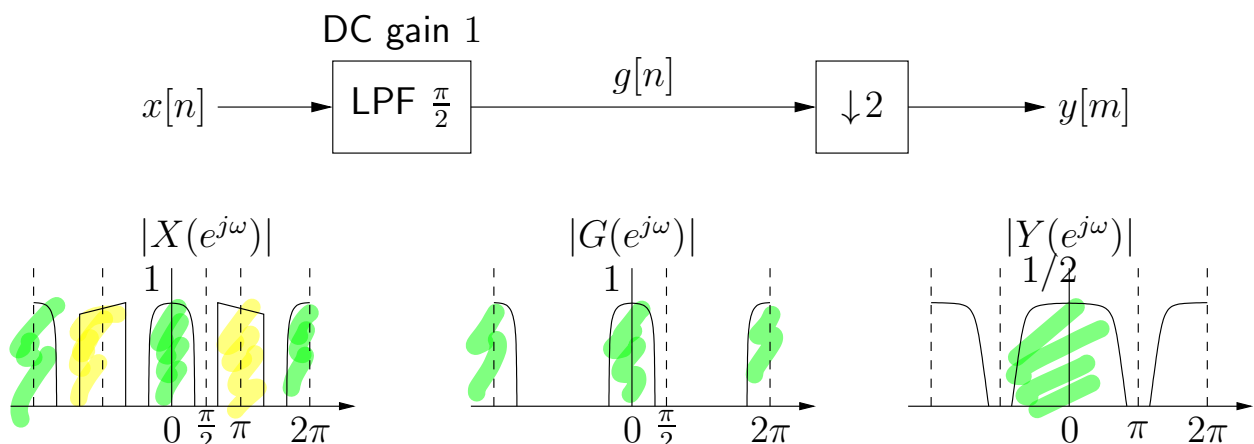
$$x(t) = g(t) + b(t),$$

good $g(t)$: lowpass and bandlimited to $\frac{1}{2DT}$ Hz

bad $b(t)$: bandpass between $\frac{1}{2DT}$ and $\frac{1}{2T}$ Hz

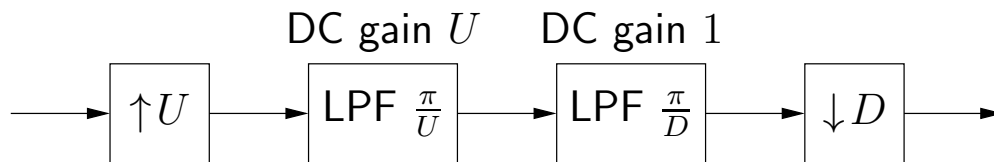
- For $\frac{1}{DT}$ -rate sampling, $g(t)$ satisfies Nyquist thm, but $b(t)$ does not, so we would have suppressed $b(t)$.
- Here, $x[n] = g[n] + b[n]$, where $b[n]$ will alias in downsampling but $g[n]$ will not. Thus, suppress $b[n]$.

- Decimation example for $D = 2$:

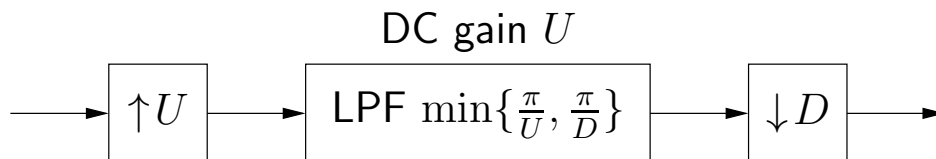


Resampling by a rational factor:

- To resample by the rational factor $\frac{U}{D}$, we first interpolate by U and then decimate by D



which can be simplified to



- Example — CD to DAT rate conversion:
 - CD sampling rate = 44.1 kHz
 - DAT sampling rate = 48 kHz.
 - Resampling ratio is $\frac{U}{D} = \frac{480}{441} = \frac{160}{147}$.
- Note:
 - The LPF filter will operate on many zero-valued input samples, and many of its outputs will be thrown away!
 - Efficient “polyphase” filter structures can avoid these redundant computations.